

Machine Learning Exercise 2B

Problem 1 *In this part of the exercise, you will implement regularized logistic regression to predict whether microchips from a fabrication plant passes quality assurance (QA). During QA, each microchip goes through various tests to ensure it is functioning correctly. Suppose you are the product manager of the factory and you have the test results for some microchips on two different tests. From these two tests, you would like to determine whether the microchips should be accepted or rejected. To help you make the decision, you have a dataset of test results on past microchips, from which you can build a logistic regression model.*

The file `ex2data2.txt` contains the dataset.

- i) Plot the data so that it displays a figure where the axes are the two test scores, and the positive and negative examples are shown with different markers.*
- ii) Feature mapping. Here you will need polynomial fitting. Define a function to map the features into all polynomial terms of x_1 and x_2 up to the sixth power. i.e:*

$$(x_1, x_2, x_1^2, x_1x_2, x_2^2, x_1^3, \dots, x_1x_2^5, x_2^6).$$

As a result of this mapping, our vector of two features (the scores on two QA tests) has been transformed into a 28-dimensional vector. A logistic regression classifier trained on this higher-dimension feature vector will have a more complex decision boundary and will appear nonlinear when drawn in our 2-dimensional plot. While the feature mapping allows us to build a more expressive classifier, it also more susceptible to overfitting, hence you will need to use regularization.

- iii) Implement the sigmoid function.*
- iv) Implement a function to calculate the cost and the gradient with regularization.*
- v) Optimise the cost function using the `minimize` function from `scipy.optimize`, using the cost and gradient functions as inputs.*
- vi) Plot the decision boundary using contour and then add the data points on top of it.*
- vii) Check how the result changes for different values of λ , try for instance $\{0, 1, 100\}$.*